SHRE HARI PREM ANANDH

AI Engineer | Production Agents, Python, LangGraph & Kubernetes on GCP

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PROFESSIONAL SUMMARY

Backend engineer who builds and runs scalable AI agent platforms in production. Built AgentForge solo in two weeks: a low-code agent platform whose LangGraph agent runs as api, worker and sweeper microservices on GKE, with a Pub/Sub queue, crash recovery and graceful shutdown. At NexCell, ships a production WhatsApp AI agent with MCP tools over Celery, Redis and PostgreSQL.

FEATURED PROJECT

AgentForge: Low-Code AI Agent Platform on Google Cloud **Live on GKE, Oct 2026**

- Built a LangGraph agent runtime on LangChain and Gemini, shipped as **3** microservices (api, worker, sweeper) from one Docker image on GKE
- Orchestrated a resilient 5-step workflow with Postgres state checkpoints, so a crashed run resumes from the last finished step
- Queued runs on Pub/Sub to leased workers that ack after saving; a sweeper re-queues stuck runs and fails poison runs after 3 tries
- Designed graceful shutdown: on SIGTERM, workers stop between steps and hand the run back; preStop draining keeps deploys zero-downtime
- Ran on GKE Autopilot with Cloud SQL via an Auth Proxy sidecar, Workload Identity and Secret Manager, autoscaling api pods from 2 to 3
- Engineered the agent harness: a Deep Agents research step with a 3-search cap and invented URLs dropped, lifting evals from **2/4** to **4/4**
- Wrote an eval harness with an LLM judge; one prompt change at a time took scores from **0/6** to **6/6**, then **15/15** on injection cases
- Enforced platform policy: Google sign-in on every request, **10** runs per user an hour, PII redaction and 30-day auto-delete, behind **128** tests

EXPERIENCE

AI Engineer Intern | NexCell Solutions **Jul 2026 - Present**

Remote, UK • Python, FastAPI, async SQLModel, Celery, Redis, PostgreSQL, Alembic, OpenAI GPT-4.1, MCP, Twilio, Docker, AWS

- Built and shipped a production WhatsApp AI agent on GPT-4.1 that calls MCP tools to qualify leads, book viewings and hand over to staff
- Developed an event-driven lead-intake pipeline on AWS SES and SNS with signature verification, phone-number deduplication and retries
- Implemented idempotent scheduled reminders across Celery and Redis workers, rolling back failed writes so no client is messaged twice
- Enforced GDPR erasure, retention and do-not-contact rules, plus AI disclosure, consent and STOP opt-outs on every WhatsApp chat
- Hardened webhooks with a separate rate-limit bucket and stripped client phone numbers and message text from worker and API logs
- Backed changes with **199** pytest files and **6** Alembic migrations across **600+** commits, merging **20+** pull requests through CI and review
- Measured production capacity on AWS EC2 and modelled LLM, messaging and hosting cost per booked lead from live conversations
- Wrote onboarding guides and reviewed pull requests for a newly joined intern taking over parts of the WhatsApp service

Software Engineer Intern | Hamiltonian Systems **Jun 2024 - Jul 2024**

Coimbatore, India • Python, FastAPI, PostgreSQL, MySQL, Grafana

- Delivered **6+** FastAPI REST endpoints with JWT authentication and Pydantic validation, powering three live client-facing applications
- Processed **100,000+** records weekly through Python pipelines and mined **50,000+** log entries to resolve **15+** recurring failures

OTHER PROJECTS

Aria: AI Voice Receptionist with an MCP Server **Deployed, Jul 2026**

- Coded a hand-rolled agent loop on gpt-4o-mini where every action runs through an MCP server over stdio, with **5** tools including booking
- Containerised the service with Docker and deployed it to Render in **2** days, with neural speech output and a browser-voice fallback

Also: MSc dissertation, a PyTorch evaluation pipeline (**64.0%** agreement against a **45.6%** null over **1,012** records); LangChain RAG service (**85%+** top-3 retrieval); IEEE disease alert system (**25%** fewer false alerts)

TECHNICAL SKILLS

Agents & GenAI: LangGraph, LangChain, Deep Agents, MCP servers and tools, tool calling, RAG, prompt versioning, evals, guardrails

Backend: Python, FastAPI, REST APIs, Pydantic, async SQLModel / SQLAlchemy, Celery, webhooks, microservices, pytest, Git

Platform & Cloud: Kubernetes (GKE), Docker, GCP (Cloud SQL, Pub/Sub, Secret Manager), AWS (EC2, SES, SNS), GitHub Actions CI/CD

Data & Messaging: PostgreSQL, Redis, Google Pub/Sub, MySQL, Alembic migrations, event-driven design, idempotent workers

Security: Google sign-in (OAuth ID tokens), JWT, Workload Identity, rate limits, PII redaction, GDPR retention, webhook signatures

Currently Learning: Go, Temporal, BigQuery, LlamaIndex

EDUCATION

MSc Artificial Intelligence | University of Sheffield, UK **Sep 2025 - Sep 2026**

Machine Learning & Adaptive Intelligence, Deep Learning, Natural Language Processing, Scalable Systems, Data Software Engineering

BE Computer Science & Engineering | Anna University (Dr NGP Institute of Technology), India **Nov 2021 - Apr 2025**

CGPA 7.89 / 10.0 • Data Structures & Algorithms, Databases, Operating Systems, Computer Networks

PUBLICATION, AWARDS & VOLUNTEERING

Publication: P. Shre Hari (first author) et al., "Real-Time Disease Monitoring and Alert System for Public Health Safety," IEEE CCPIS, 2025

Certifications: Claude Code 101 (LinkedIn, 2026) • Prompt Engineering for LLMs (Coursera) • AI & Cybersecurity: Anomaly Detection

Awards: Smart India Hackathon Finalist, 2023 • First Place, Extempore, World Space Week 2023 (ISRO) • Class Representative, 60+ cohort

Leadership: Directed a 4-member team with no missed deadlines • Delivered 5 technical sessions, including a TED-style talk

Volunteering: Monthly UNICEF India donor (2023 to 2025) • Ran a UNICEF campaign with 3 aid camps and 2 blood drives (250+ people)

Languages & Memberships: English (fluent), Tamil (native), Japanese (basic) • ISTE and the Computer Society of India (CSI)

Interests: agent platforms, evaluation of generative models, AI coding tools, developer tooling, open source